Ex 1-5

1. $3x + 7 = -8$

   $3x = -15$

   $x = -5$
2. $2x + 3 = x - 1$

   $2x + 4 = x$

   NS
3. $5x + 3 = 2x + 3$

   $3x + 3 = 3$

   $3x = 3$

   $x = 0$
4. $4x + 3 = x - 8$
   1. {rational only}

      $4x = x - 11$

      $3x = -11$

      $x = \frac{-11}{3}$
   2. {ints}

      NS
5. $x^2 = 16$
   1. {neg only}

      $\sqrt{x^2} = \sqrt{16}$

      $x = 4$

      $4 > 0$

      NS
   2. {real only}

      $x = 4$
6. $x^2 = \frac{1}{9}$
   1. {rational only}

      $\sqrt{x} = \sqrt{\frac{1}{9}}$

      $x = 0.\overline{333}$
   2. {ints}

      NS
7. $(x + 3)(3x - 2) = 0$

   $x + 3 = 0$

   $x = -3$

   $3x - 2 = 0$

$3x = 2$

$x = \frac{2}{3}$

8. $(2x - 5)(x + 1) = 0$

$x + 1 = 0$

$x = -1$

$2x - 5 = 0$

$2x = 5$

$x = \frac{5}{2}$

9. $(x + 3)(3x - 2) = 0$

$x + 3 = 0$

$x = -3$

$3x - 2 = 0$

$3x = 2$

$x = 2/3$

10. $(2x + 3)(x - 5)(x + 1) = 0$

$2x + 3 = 0$

$2x = -3$

$x = \frac{-3}{2}$

$x - 5 = 0$

$x = 5$

$x + 1 = 0$

$x = -1$

11. $x(2x - 1)(x + 4) = 0$

$x = 0$

$2x - 1 = 0$

$2x = 1$

$x = \frac{1}{2}$

$x + 4 = 0$

$x = 4$

12. $|x| = 7$

$x = 7$ or $x = -7$